

## EXERCISE 6: INTENSITY DISCRIMINATION AND BINAURAL INTERACTION

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### 1. INTRODUCTION

#### 1.1. Exercise Objectives

After completion of this lab exercise and the associated report it is expected that you should be able to:

- State what JND, ITD, ILD, and BMLD stand for and explain what they mean
- Relate intensity discrimination to Weber's law
- Explain why a binaural processor must exist
- Explain what is meant by "time-intensity trading"
- Derive the trading ratio from measurements
- Discuss the relationship between trading ratio and level

#### 1.2. Some Notes before Starting

Several measurements will be carried out during this exercise. It will take each subject about one hour to complete all measurements. If all group members perform all measurements, there will be very little time left for analysis and discussion of the results. Therefore, if time runs out, it is recommended that one group member measures all of the data points, whereas other group members only perform one repetition of the experiment. This will give everyone the idea of what the different stimuli sound like, and what the detection cues may be.

A "ready-to-copy" version of the cited articles and book chapters can be found in the course reader or in the supplementary reader in the library. More detailed background information and further references can be found in nearly all hearing textbooks, e.g., [1] or [2].

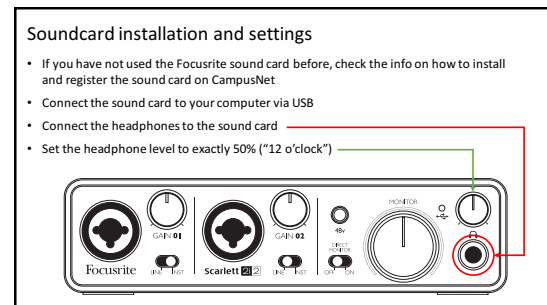
#### 1.3. Exercise setup

For this exercise, the following statements should be entered after starting MATLAB in order to be able to access the provided scripts:

```
startup % sets the path
```

There might not be enough sound cards for all groups, so it's perfectly fine to use your computer's built-in audio output instead. If you do get a Focusrite sound card, please

verify that the *sound card level is set to 50%*. If not already done, connect the sound card to the computer and install the drivers. Afterwards, please connect the headphones (Sennheiser HD 380 pro) to the sound card.



### 2. PART 1: JUST NOTICEABLE INTENSITY DIFFERENCE

#### 2.1. Monaural JND

One of the most studied properties of the auditory system is its ability to discriminate among sounds differing only in intensity. The smallest difference a listener can detect is called "Just Noticeable Difference" or, in short JND. A JND does not refer to a particular unit. Regarding an intensity discrimination task, one refers to the smallest detectable change in intensity as the "intensity JND". According to Weber's law, the intensity JND,  $\Delta I$ , is proportional to the sound intensity,  $I$ , expressed in  $W/m^2$ :

$$\frac{\Delta I}{I} = K \quad (1)$$

The constant  $K$  in Eq. 1 is called the "Weber fraction". The intensity JND is more commonly expressed as a level JND in decibel (dB), referred to as  $\Delta L$ . It can be expressed as:

$$\Delta L = 10 \cdot \log_{10}(1 + K) \quad (2)$$

##### 2.1.1. Experiment

In the first experiment, you will measure your individual intensity JND for a broadband noise (250-4000 Hz, 500 ms) at two different presentation levels (65 and 75 dB SPL), using an adaptive three-interval, three-alternative, forced-choice procedure with a one-up, two-down tracking rule. JNDs will

be measured **twice for each level**, resulting in a total of 4 measurements. The experiment is provided within the MATLAB AFC framework and is called `intjnd`. Please have a look at Sec. A.1 on how to start an experiment and Sec. A.2 on how to plot and access data. No user parameters are needed. In the experiment, you will have to indicate the interval which differs from **(is louder than) the other ones**. (Experiment duration: approximately **8 minutes per listener**)

### 2.1.2. Questions

- **[BACKGROUND]** Derive Eq. (2) for the level JND in decibel. *Hint:*  $\Delta L = L_2 - L_1$
- **[DISCUSS]** What results do you expect for the different levels, keeping Weber's law in mind? Do your results align with the expectation? If no, why could that be?
- **[RESULTS]** The results of the measurement are given as level JNDs  $\Delta L$ . Create an additional plot with the Weber fraction as a function of the sound level.

## 2.2. Interaural JND

In the last experiment the intensity JND was measured monaurally. In this experiment, the **JND for interaural level differences (ILD)** will be measured using the 65 dB SPL stimulus from the previous experiment as a basis to create the dichotic stimuli. The ILD can be created using one of the following two methods:

- A:** Increase the level in one ear by the ILD value while the level in the other ear is unchanged.
- B:** Increase the level in one ear by half the ILD value and decreasing the level in the other ear by half the ILD value.

### 2.2.1. Experiment

In the experiment, the stimulus in the signal interval will be presented with the sound level in the right ear lowered by half the interaural level difference, and the sound level in the left ear raised by half the interaural level difference (**Method B**). This experiment is also provided within the MATLAB AFC framework and is called `ildjnd`. Please refer to Sec. A.1 for instructions on starting an experiment and Sec. A.2 on plotting and accessing the data. All parameters describing the experiment are the same as in the previous experiment, except the interaural difference explained above. No user parameters are needed. In the experiment you will have to **indicate the interval which is localized to the left of the other ones**. (Experiment duration: approximately **4 minutes per listener**)

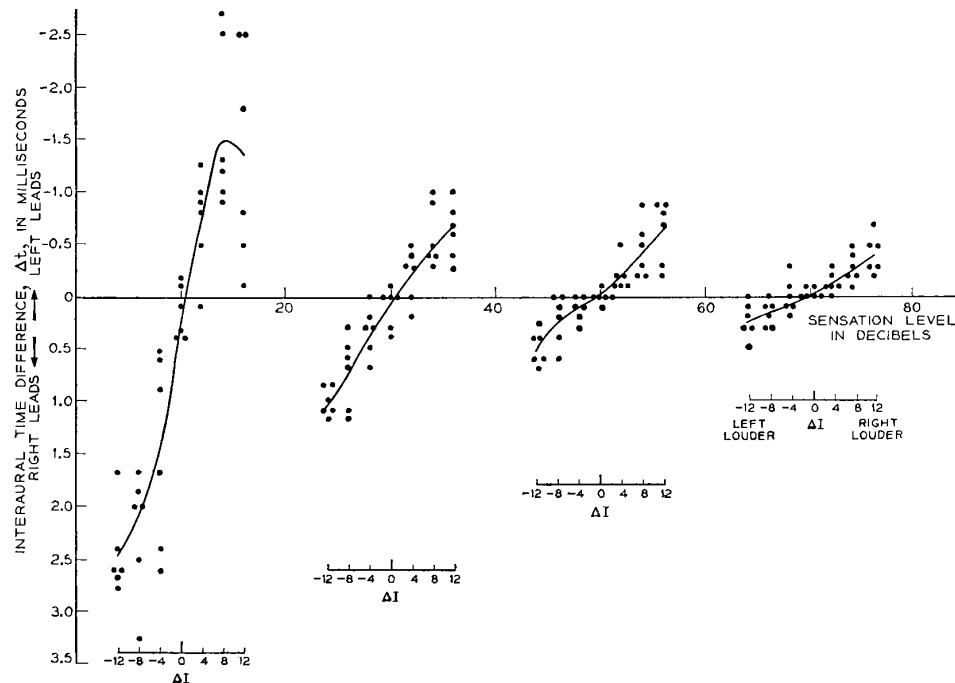
### 2.2.2. Questions

#### [DISCUSS]

- Assuming that we do *not* have the ability to combine the left and right ear signals in our auditory system (no “binaural processor”), how are the JNDs for ILDs expected to relate to the monaural level JND measured in the previous experiment, if Method A is used to create the ILD, and if Method B is used to create the ILD?
- Compare your data with those expected under the assumption that one does not have a binaural processor.
- Compare the data with those from the previous experiment. Does your data “prove” that there is a binaural processor?
- How did you perceive the noise stimuli? Did you hear two independent sound objects or one fused object in each interval? Describe your perception of the noise stimuli.

## 3. PART 2: TIME-INTENSITY TRADING

Differences between the ear input signals are necessary in order for an auditory object to appear away from the median plane. There are **two main physical differences between the sounds that arrive at the two ear drums: Interaural Level Differences (ILD) and Interaural Time Differences (ITD)**. Either of these differences give rise to a perceived lateralization, or shifting of the sound image away from the median plane. **When presented over headphones, a sound with an ILD will be lateralized within the head toward the ear with the higher presentation level. Similarly, a sound with an ITD will be lateralized toward the leading ear. The relative perceptual strength of the two interaural differences can be investigated in an experiment where the laterality obtained with ITDs and ILDs in opposite direction is studied. Such experiment is called “trading” experiment. It provides a measure of the ITD that is needed to compensate for the laterality created by an ILD. The ratio between ITD and ILD determined in a trading experiment, measured in  $\mu\text{s}/\text{dB}$ , is called the “compensation factor” or “trading ratio.” It has been reported in the literature that this ratio depends on sound level [3]. When signals are louder, smaller ITDs are required to compensate for a given ILD. An often-quoted illustration of this dependence is shown in Fig. 1. The phenomenon of time-intensity trading led to the theory that time and intensity differences may be coded in a similar way in the auditory system. However, separate sound images or a spatially broad image may be heard when time and intensity are opposed. This indicates that time and intensity are not truly equivalent.**



**Fig. 1.** Trading curves of a typical subject for broadband clicks at various levels. The curves drawn through the plots were obtained using third-order polynomial fits [3, Fig. 3].

### 3.1. Experiment

Your next measurement is such a trading experiment. The setup is rather simple: An adaptive two-interval, two-alternative, forced-choice procedure with a one-up, one-down tracking rule is used. The stimuli presented in the two intervals of a trial are identical, except that the stimuli to the ears are being swapped between the intervals. The stimulus is a binaural click train consisting of ten five-sample clicks separated by 2200 samples. The sampling frequency is 44.1 kHz. Half of the ILD is applied as a gain in one channel of the binaural stimulus, and half as attenuation in the other channel. The tracking variable, the ITD, is similarly divided between the two channels, with one channel shifted forward by half the ITD and the other shifted backward. The task during the measurement is to indicate the direction of motion between the intervals (i.e. the second interval is localized to the right or to the left of the first interval). The ITD is measured for seven different ILDs (-10, -7, -4, 0, 4, 7, and 10 dB). Each point is measured twice, resulting in a total of 14 measurements. To start the experiment `TItrade` you have to provide the level as the user parameter. The time intensity trade can be done at different levels up to 60 dB SPL. You should at least measure two different levels (e.g., 60 dB SPL and 45 dB SPL). In the experiment you will have to indicate the direction of motion between the intervals. To represent the data, use the `pldata` command with the additional user parameter. The ITD data will be given in samples. Do not forget to convert it

to ms! (Experiment duration: approximately 10 minutes per listener per level)

### 3.2. Questions

[DISCUSS]

- Why is a “time-intensity-traded” stimulus not natural?
- How do you perceive the “unnatural” stimuli? Do you have a clear lateralization in all cases? Did you hear two separate sound images?
- In most cases the data are not appropriately described by a linear function. Therefore, it is quite common to model the data with a third-degree polynomial [3]. Use the MATLAB function `polyfit` to fit a polynomial to your data (type `help polyfit` for usage details). Plot the fitted polynomial together with the data.
- Normally we would expect the data to be symmetrical about the origin. However, some kind of bias towards one side is quite common. The zero-order term of the polynomial indicates the bias of your data. What could cause such bias?
- The first-order coefficient of the fitted polynomial gives the trading ratio. How does the trading ratio change with level? (see also Fig. 1)

#### 4. PART 3: BINAURAL MASKING LEVEL DIFFERENCE (BMLD)

In the previous experiments we examined the auditory system's sensitivity to changes in interaural level and time difference, which are principally used to locate sound sources in the horizontal plane. The ability to perceive interaural differences is also observed in masking experiments. Many studies have shown that the threshold for detecting a signal masked by noise is lower when the signal and the noise are presented with different interaural properties. For instance, for a noise presented identically to both ears (diotic), the masked thresholds is about 15 dB lower if the signal is presented to both ears with a 180 interaural phase difference than if the signal is presented diotically.

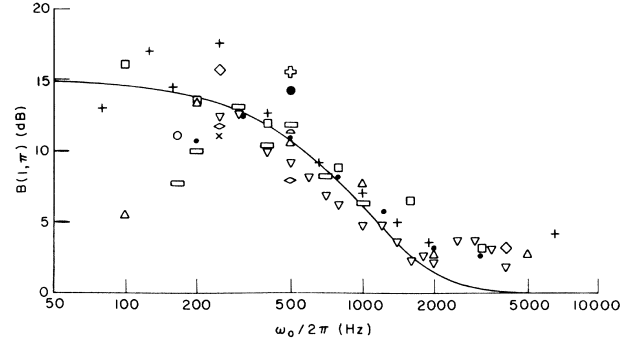
The difference between these two masked-threshold levels is known as the binaural masking-level difference, abbreviated as "BMLD". An abbreviation was introduced to describe these types of experiments with the symbols "N" (for noise) and "S" (for signal), each followed by a suffix denoting the interaural phase relation. E.g., a phase difference of 180 (or  $\pi$  in radians) for the signal is denoted by  $S_\pi$ . Thus,  $N_0S_0$  refers to the condition where both noise and signal have identical phase at both ears, and  $N_0S_\pi$  refers to the condition where the left and right noises are in phase, but the signals are inverted in phase. A very detailed discussion of different parameters affecting the BMLD can be found in [4]. In this exercise we will focus on the BMLD as a function of the signal frequency.

A model that is able to describe the BMLD is the "equalization-cancellation model" (EC model, [5, 6]). The basic idea of this model is that the auditory system tries to eliminate the masker by manipulating the stimuli between the two ears until the masking components are the same (equalization), and then subtracting the signals (cancellation). The equalization process amplifies or attenuates the stimuli to compensate for an ILD and time-shifts the stimuli to compensate for an ITD. Depending on the masker and signal configuration, the signal/masker ratio could be infinitely improved after the cancellation process, i.e., when the masker is entirely cancelled. In order to obtain only finite improvements, the model assumes an imperfect equalization process by introducing random jitter in the amplification ( $\epsilon$ ) and time shifting ( $\delta$ ).  $\epsilon$  and  $\delta$  are assumed to be statistically independent random variables with zero mean and variances  $\sigma_\epsilon^2$  and  $\sigma_\delta^2$ . Using these (and other) assumptions, [5] was able to show that the difference between the  $N_0S_0$  and  $N_0S_\pi$  condition can be described by the following equation:

$$L(N_0S_0) - L(N_0S_\pi) = 10 \cdot \log_{10} \left( 1 + \frac{2 \cdot e^{-\omega_0^2 \sigma_\delta^2}}{1 + \sigma_\epsilon^2 - e^{-\omega_0^2 \sigma_\delta^2}} \right), \quad (3)$$

where  $\omega_0$  is the center frequency (in radians) of the auditory filter with the stimulus. In our case, this can be assumed to be

equal to the frequency of the sinusoid.



**Fig. 2.** BMLD ( $N_0S_0$ - $N_0S_\pi$ ) as a function of  $\omega_0$ . Data points are from 18 different studies. The curve is from Eqn. 3 with  $\sigma_\delta = 105 \mu s$  and  $\sigma_\epsilon = 0.25$ . For more details, see [6, Fig. 4].

##### 4.1. Experiment

In this experiment, the signals are 480-ms sinusoids with an interaural phase difference of 0 or 180. Signal frequencies of 250, 500, 1000 and 2000 Hz will be used during the measurement. The masker is a 500-ms broadband noise (2 octaves wide, 65 dB SPL), geometrically centered around the signal frequency. The presentation of the masker starts 10 ms before and ends 10 ms after the presentation of the signal. The same masker will be presented at both ears. The thresholds in both the  $N_0S_0$  and  $N_0S_\pi$  conditions will be measured. The masked threshold will be measured using an adaptive three-interval, three-alternative, forced-choice procedure with a one-up, two-down tracking rule. Each point will be measured twice, resulting in a total of 16 measurements (2 conditions, 4 frequencies, each twice). The experiment is provided within the MATLAB AFC framework with the name BMLD. No user parameters are needed. Again, please have a look at Sec. A.1 on how to start the experiment. In the experiment you will have to detect the pure-tone signal in the noise. (Experiment duration: approximately 30 minutes per listener). If you run out of time, you may stop the measurement after 8 tracks (press 'e' instead of 's') and let another group member perform another set of 8 tracks.

In this experiment we are not interested in the absolute masking level, but in the difference of masking levels. Therefore, a new function `p1BMLD` is provided to be able to read out and plot the BMLD:

```
[vFreq, vBMLD, vStd] =  
    p1BMLD(vp, group[, proc])
```

The input parameters are almost the same as the ones used by the function `p1data`:

- **vp:** The initials of the subject who did the measurement. To see results of more than one subject, enclose the initials with curly brackets. {}. Examples:
  - 'ay'
  - {'ay', 'bm', 'co'}
- **group:** The name of your group
- **proc:** Optional Parameter. Possible values:
  - 'mean' Plot the mean BMLD (default)
  - 'raw' Plot the raw data for each condition

The output parameters are

- **vFreq:** Frequencies used in the experiment
- **vBMLD:** Mean BMLD in dB ( $N_0S_0 - N_0S_\pi$ )
- **vStd:** Standard deviation of BMLD

## 4.2. Questions

- **[BACKGROUND]** Use Eqn. 3 to predict the BMLD for very high and for very low frequencies. Which variable dominates the BMLD values in these situations? Can you explain this?
- **[RESULTS]** Do the actual measurement and compare your data with the data given in Fig. 2. Only include the 'mean' data unless the 'raw' data are needed to explain some unexpected results.
- **[IMPLEMENT]** Implement Eqn. 3 as a function `ECmodel`. A header for the function is supplied. Type `edit ECmodel` to open the corresponding m-file. Remember to implement it in a vectorized way, such that you are able to give a vector of frequencies to the function to get a vector of the corresponding BMLDs. You can easily check your implementation by comparing its output with the curve in Fig. 2 using  $\sigma_\delta = 105 \mu\text{s}$  and  $\sigma_\epsilon = 0.25$ .
- **[IMPLEMENT]** Fit the model to your data using the function `ECfit`. This function uses your implementation of `ECmodel`.
- **[DISUCSS]** Compare your fitted model to literature.

## 5. REFERENCES

- [1] Brian C. J. Moore, *An Introduction to the Psychology of Hearing*, Academic Press, 5th edition, 2003.
- [2] William A. Yost, *Fundamentals of Hearing – An Introduction*, Academic Press, San Diego, 2000.
- [3] E. E. David, Newman Guttman, and W. A. van Bergeijk, "Binaural interaction of high-frequency complex stimuli," *J. Acoust. Soc. Am.*, vol. 31, no. 6, pp. 774–782, 1959.
- [4] David M. Green and William A. Yost, "Binaural analysis," in *Physiology (CNS) Behavioural Studies Psychoacoustics*, Wolf D. Keidel and William D. Neff, Eds., vol. V/2 of *Handbook of Sensory Physiology*, pp. 461–480. Springer-Verlag, Berlin, 1975.
- [5] Nathaniel I. Durlach, "Equalization and cancellation theory of binaural masking-level differences," *J. Acoust. Soc. Am.*, vol. 35, no. 8, pp. 1206–1218, 1963.
- [6] Nathaniel I. Durlach, "Binaural signal detection: Equalization and cancellation theory," in *Foundations of Modern Auditory Theory*, Vol. 2, Jerry V. Tobias, Ed., pp. 369–462. Academic Press, New York, 1972.

## A. AFC MATLAB TOOLS

### A.1. Starting measurements

To start a measurement you have to use the `afc_main` command:

```
afc_main(expname, vp, [userpar,] group);
```

The input parameters are as follows:

- `expname`: The name of the experiment
- `vp`: The initials of the subject
- `userpar`: An optional user parameter. If an experiment does not require user parameters, just omit this. Read the instructions of the experiments for more details.
- `group`: Your group number

For example, to start the experiment “intjnd” for subject “Anon Ymous”, member of group 3, enter the following command:

```
afc_main('intjnd','ay','group3'); % This is just an example!
```

Another example: start the experiment “Titrade” for subject “Unk Nown”, member of group 5 with a level of 60 dB SPL:

```
afc_main('Titrade','un','60','group5'); % This is just an example!
```

Use the number pad to answer the questions during the measurement. Make sure that you activated “NumLock” (check the NumLock-LED).

### A.2. Plotting the results

The data of all the measurements will be saved in your current working directory<sup>1</sup>. The different filenames follow one of the schemes

```
expname_vp_group.dat  
expname_vp_userpar_group.dat
```

Thus, the data for the examples above could be found in the files `intjnd_ay_group3.dat` and `Titrade_un_60_group3.dat`. To plot the results of AFC measurements, use the command `pldata`:

```
[x,y,ystd] = pldata(expname, vp, [userpar,] group, proc);
```

The input parameters are almost the same as described above:

- `expname`: The name of the experiment
- `vp`: The initials of the subject who did the measurement. To see results of more than one subject, enclose the initials with curly brackets. { }. Examples:
  - `'ay'`
  - `{ 'ay', 'bm', 'co' }`
- `userpar`: The optional user parameter
- `group`: The name of your group
- `proc`: Optional Parameter. Possible values:
  - `'mean'` Plot the mean data
  - `'raw'` Plot the raw data (default)

The (optional) output parameters are:

- `x`: The experimental parameter
- `y`: The results of the experiment
- `ystd`: The standard deviation of y

You can use the outputs of `pldata.m` to access your data.

<sup>1</sup>If you changed the directory (e.g., by using the `cd` command) you can get the actual working directory with the `pwd` command.